

# Advanced Programming Practices

## MTech 2026

Lab - 1

---

### General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo\_APP1\_Qn.c (e.g. ABC2026001\_APP1\_Q1.c).
  - Every program must compile without warnings using the gcc compiler.
  - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
  - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
  - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
- 

1. Write a program to scan an integer array once and returns both the second-largest and second-smallest distinct values via pointer, handling arrays that are too small or contain duplicates only.
2. Given an array of 0s, 1s, and 2s, rearrange it in a single pass so all 0s come first, then all 1s, then all 2s, without using a counting/sorting library call. Time complexity should be  $O(n)$  and space complexity should be  $O(1)$ .
3. Given two sorted integer arrays A (size m, with m+n capacity) and B (size n), merge B into A in-place so A ends up sorted, without using extra array space.
4. Given a matrix with many zero entries, store only its non-zero elements as a list of (row, col, value) triples, print this compact representation, and write a function that reconstructs and prints the original matrix from it.
5. Find all elements of a matrix that are simultaneously the minimum in their row and the maximum in their column (matrix saddle points), or report that none exist.